

Step 3: Sequence Generation

Continuous 3D Morphology Synthesis

The Objective

Once the Backward Diffusion model predicts that two fragments connect, the **Sequence Generator** predicts how.

It generates a continuous sequence of 3D coordinates $X_{\{1:L\}}$ that bridges the topological gap, conditional on the invariant contexts of the two fragments.

The Denoising Process (DDPM)

The model functions as a continuous Denoising Diffusion Probabilistic Model. - **Input:** Pure Gaussian noise $N(0, I)$ sampled into a 3D tensor of shape (Batch, MaxLength, 3). - **Conditioning:** The invariant Context Vectors (c_1, c_2) extracted from the Backward Diffusion Model. - **Output:** A smooth, biologically plausible 3D curve tracing a neurite path.

Layer-wise Architecture: 1D-Chain EGNN

```
graph TD
    X[Noisy Coords N, 3] --> EGNN[1D-Chain EGNN]
    T[Time Embed] --> SumH[Sum Features]
    Ctx[Context Embed] --> SumH
    Pos[Pos Embed] --> SumH
    SumH --> EGNN

    EGNN --> Noise[Noise Prediction]
    EGNN --> Type[Node Type Logits]
```

Optimization: Morphological Losses

The generation is optimized using two primary losses: - **Coordinate Loss (Weight: 10.0):** Masked Mean Squared Error (nn.MSELoss) over the predicted coordinate noise vs the added Gaussian noise. The heavy weight prioritizes smooth, accurate paths. - **Node Type Loss (Weight: 0.5):** Masked Cross-Entropy Loss for structural intents.

Padding masks ensure that variable-length branches do not corrupt the loss gradient.

Example: Generated 3D Sequences

Generated Morphologies